

8086 Microprocessor & Pin Diagram

Sk. Shalauddin Kabir

Lecturer,

Dept. of Computer Science and Engineering,

Jashore University of Science and Technology.

Microprocessor

Program controlled semiconductor device (IC) which fetches (from memory), decodes and executes instructions.

It is used as CPU (Central Processing Unit) in computers.

Microprocessor

Third Generation

During 1978

HMOS technology $\Rightarrow$ Faster speed, Higher packing density

16 bit processors $\Rightarrow$ 40/ 48/ 64 pins

Easier to program

Dynamically relatable programs

Processor has multiply/ divide arithmetic hardware

More powerful interrupt handling capabilities

Flexible I/O port addressing

Intel 8086 (16 bit processor)

First Generation

Between 1971 – 1973

PMOS technology, non compatible with TTL

4 bit processors $\Rightarrow$ 16 pins

8 and 16 bit processors $\Rightarrow$ 40 pins

Due to limitations of pins, signals are multiplexed

Fifth Generation **Pentium**

Fourth Generation

During 1980s

Low power version of HMOS technology (HCMOS)

32 bit processors

Physical memory space 2^{24} bytes = 16 Mb

Virtual memory space 2^{40} bytes = 1 Tb

Floating point hardware

Supports increased number of addressing modes

Intel 80386

Second Generation

During 1973

NMOS technology $\Rightarrow$ Faster speed, Higher density, Compatible with TTL

4 / 8/ 16 bit processors $\Rightarrow$ 40 pins

Ability to address large memory spaces and I/O ports

Greater number of levels of subroutine nesting

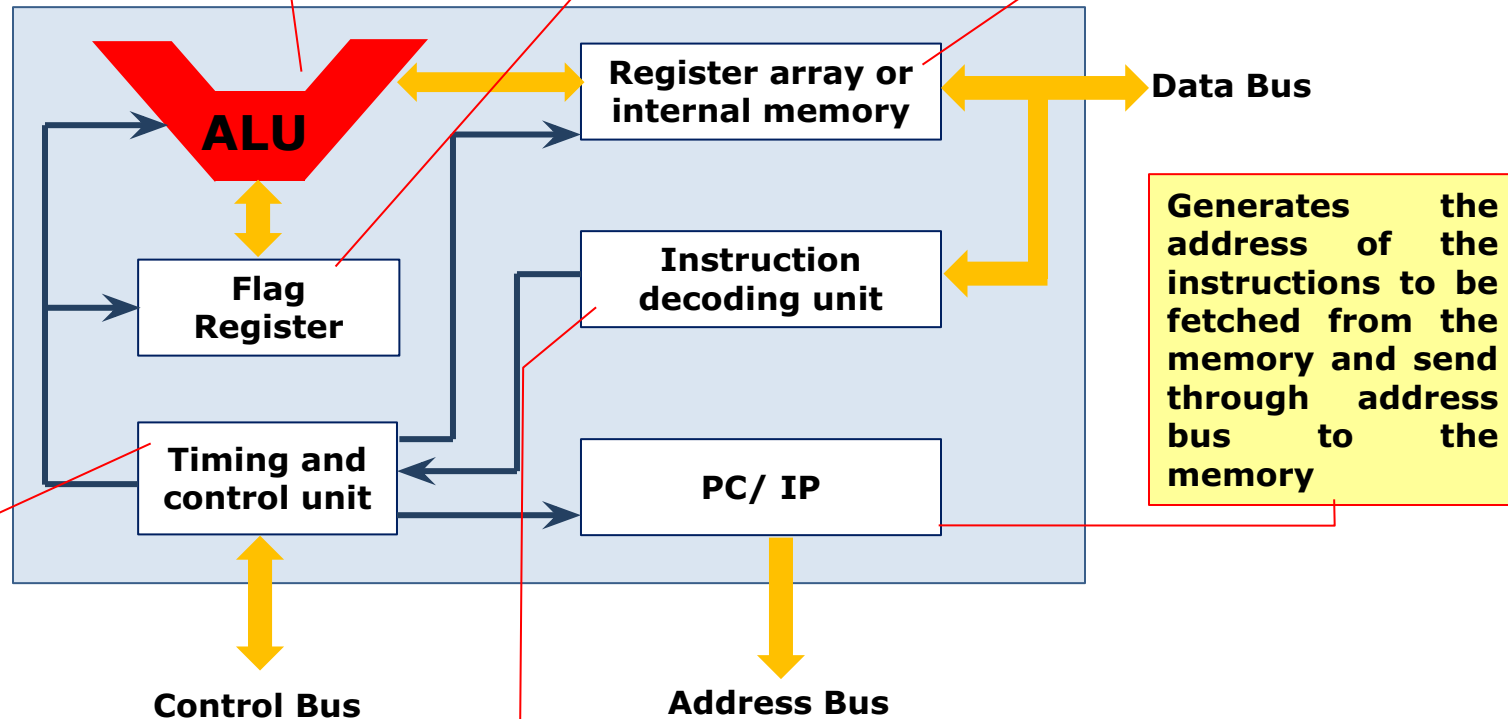
Better interrupt handling capabilities

Intel 8085 (8 bit processor)

Computational Unit;
performs arithmetic and logic operations

Various conditions of the results are stored as status bits called flags in flag register

Internal storage of data



Generates control signals for internal and external operations of the microprocessor

Decodes instructions; sends information to the timing and control unit

Generates the address of the instructions to be fetched from the memory and send through address bus to the memory

First 16-bit processor released by INTEL in the year 1978

Originally HMOS, now manufactured using HMOS III technique

Approximately 29,000 transistors, 40 pin DIP, 5V supply

Does not have internal clock; external asymmetric clock source with 33% duty cycle

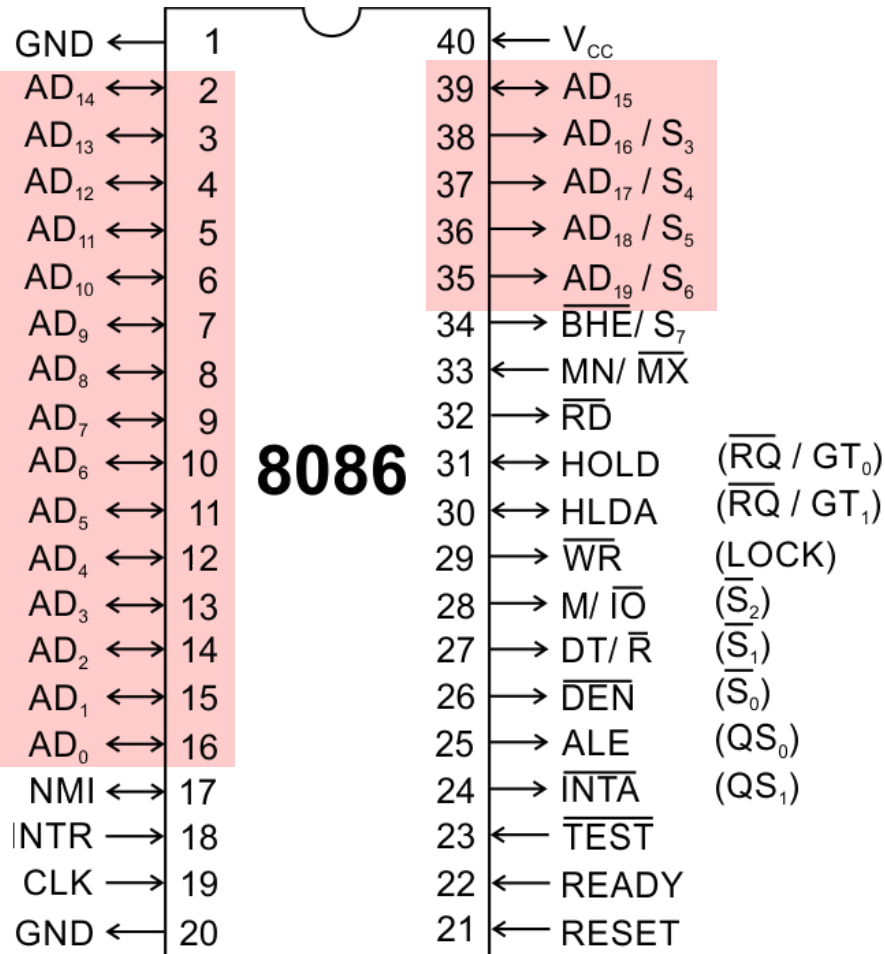
20-bit address to access memory $\Rightarrow$ can address up to $2^{20} = 1$ megabytes of memory space.

Addressable memory space is organized into two banks of 512 kb each; **Even (or lower) bank** and **Odd (or higher) bank**. Address line A_0 is used to select even bank and control signal $\overline{BHE}$ is used to access odd bank

Uses a separate 16 bit address for I/O mapped devices $\Rightarrow$ can generate $2^{16} = 64$ k addresses.

Operates in two modes: **minimum mode** and **maximum mode**, decided by the signal at $\overline{MN}$ and $\overline{MX}$ pins.

Pins and signals



AD_0-AD_{15} (Bidirectional)

Address/Data bus

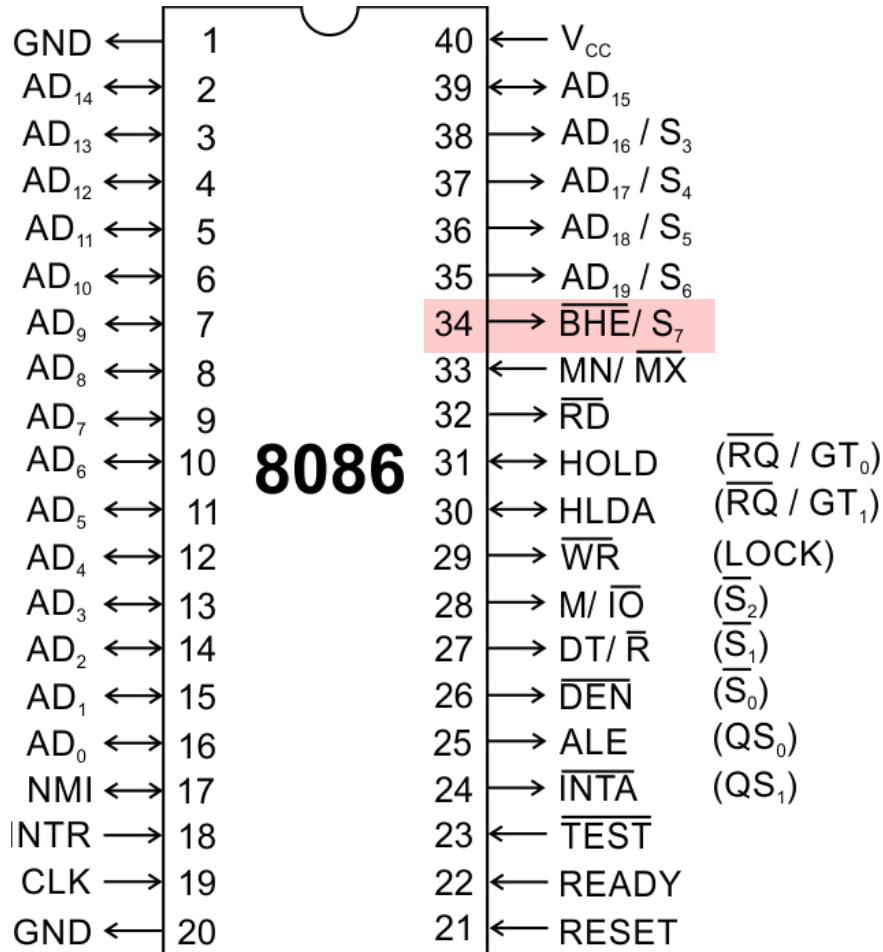
Low order address bus; these are multiplexed with data.

When AD lines are used to transmit memory address the symbol A is used instead of AD, for example A_0-A_{15} .

When data are transmitted over AD lines the symbol D is used in place of AD, for example D_0-D_7 , D_8-D_{15} or D_0-D_{15} .

$A_{16}/S_3, A_{17}/S_4, A_{18}/S_5, A_{19}/S_6$

High order address bus. These are multiplexed with status signals



BHE (Active Low)/ S_7 (Output)

Bus High Enable/Status

It is used to enable data onto the most significant half of data bus, D_8 - D_{15} . 8-bit device connected to upper half of the data bus use BHE (Active Low) signal. It is multiplexed with status signal S_7 .

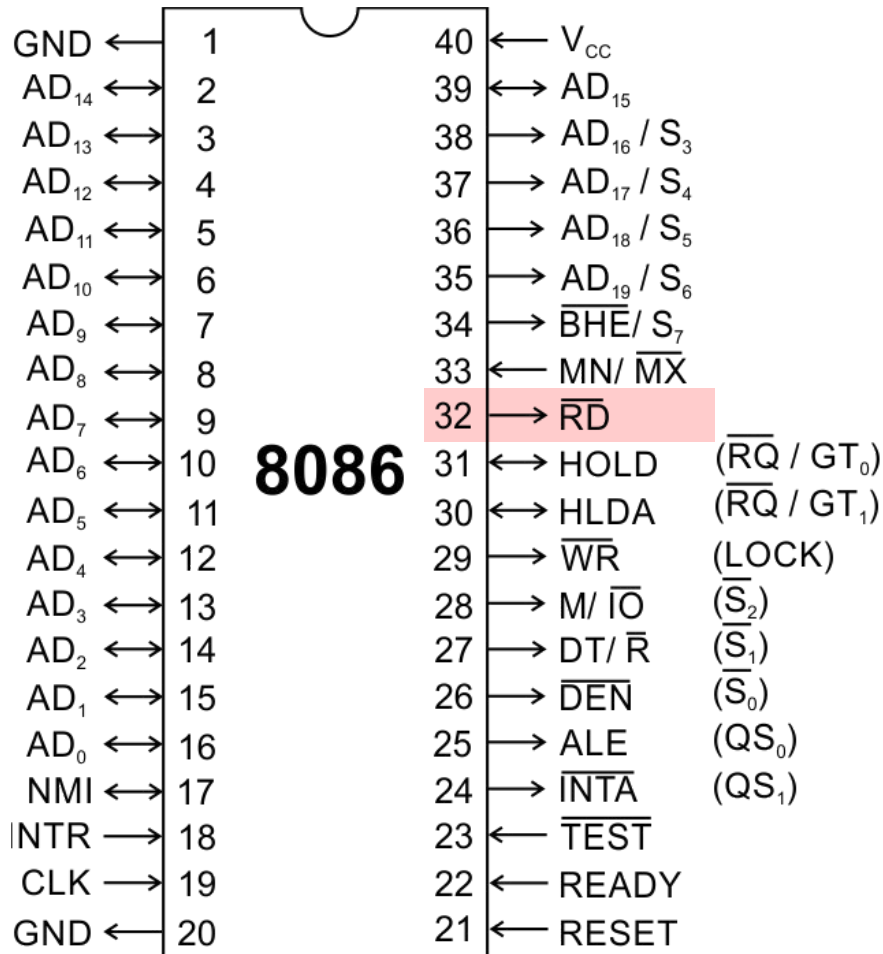
MN/ $\overline{MX}$

MINIMUM / MAXIMUM

This pin signal indicates what mode the processor is to operate in.

RD (Read) (Active Low)

The signal is used for read operation.
It is an output signal.
It is active when low.



TEST

$\overline{TEST}$ input is tested by the 'WAIT' instruction.

8086 will enter a wait state after execution of the WAIT instruction and will resume execution only when the $\overline{TEST}$ is made low by an active hardware.

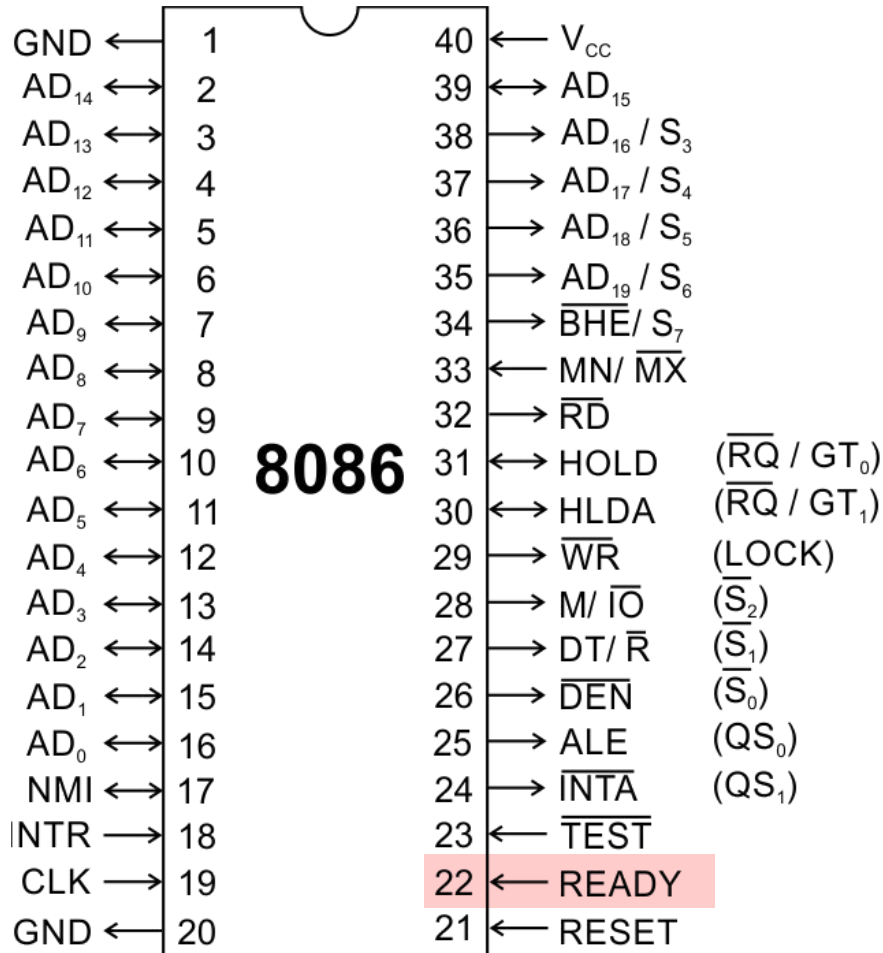
This is used to synchronize an external activity to the processor internal operation.

READY

This is the acknowledgement from the slow device or memory that they have completed the data transfer.

The signal made available by the devices is synchronized by the 8284A clock generator to provide ready input to the 8086.

The signal is active high.



RESET (Input)

Causes the processor to immediately terminate its present activity.

The signal must be active HIGH for at least four clock cycles.

CLK

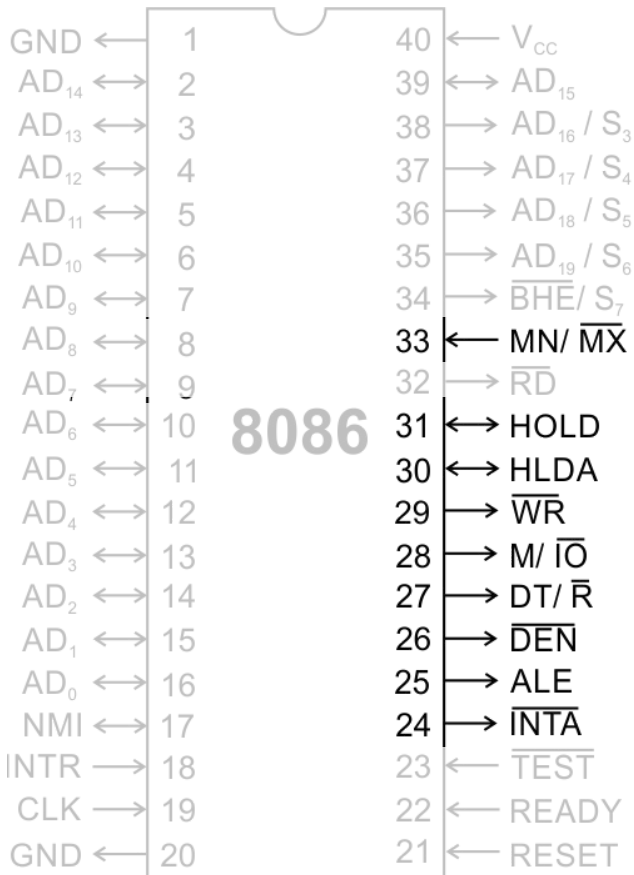
The clock input provides the basic timing for processor operation and bus control activity. Its an asymmetric square wave with 33% duty cycle.

INTR Interrupt Request

This is a triggered input. This is sampled during the last clock cycles of each instruction to determine the availability of the request. If any interrupt request is pending, the processor enters the interrupt acknowledge cycle.

This signal is active high and internally synchronized.

The 8086 microprocessor can work in two modes of operations : **Minimum mode** and **Maximum mode**.

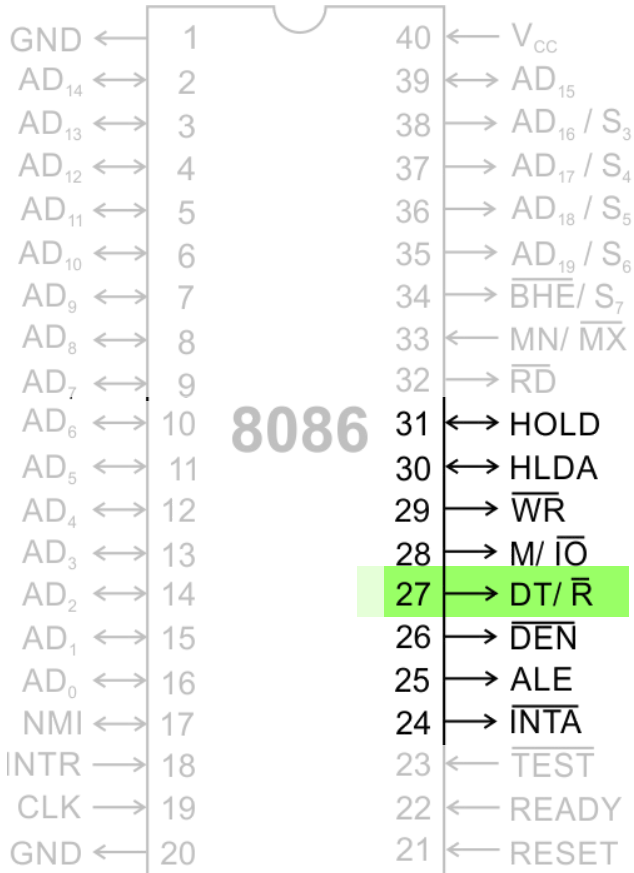


In the minimum mode of operation the microprocessor do not associate with any co-processors and can not be used for multiprocessor systems.

In the maximum mode the 8086 can work in multi-processor or co-processor configuration.

Minimum or maximum mode operations are decided by the pin MN/ MX(Active low).

When this pin is high 8086 operates in minimum mode otherwise it operates in Maximum mode.



Pins 24 -31

For minimum mode operation, the MN/ $\overline{\text{MX}}$ is tied to VCC (logic high)

8086 itself generates all the bus control signals

(**Data Transmit/ Receive**) Output signal from the processor to control the direction of data flow through the data transceivers

(**Data Enable**) Output signal from the processor used as out put enable for the transceivers

ALE (**Address Latch Enable**) Used to demultiplex the address and data lines using external latches

Used to differentiate memory access and I/O access. For memory reference instructions, it is **high**. For IN and OUT instructions, it is **low**.

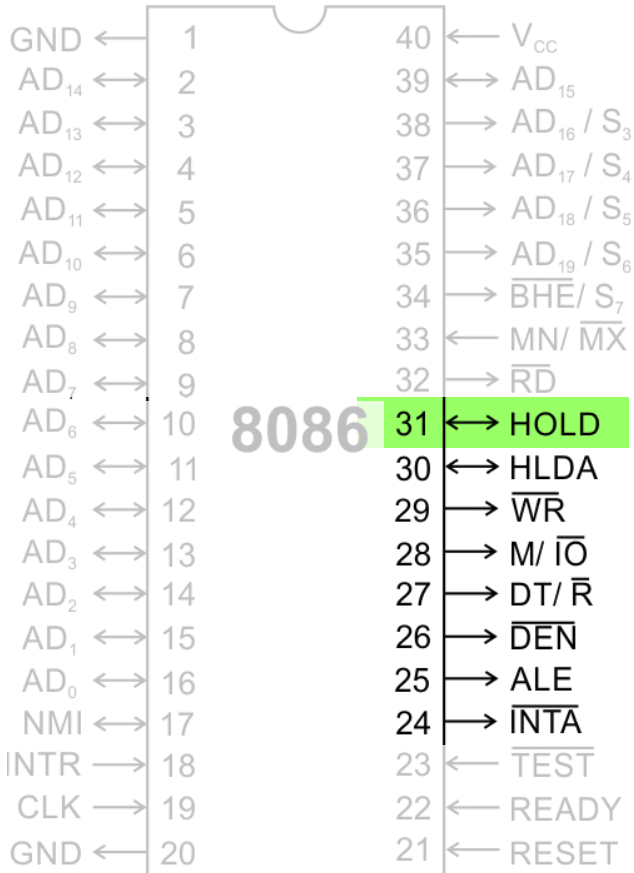
Write control signal; asserted **low** Whenever processor writes data to memory or I/O port

(**Interrupt Acknowledge**) When the interrupt request is accepted by the processor, the output is **low** on this line.

Pins 24 -31

For minimum mode operation, the $\text{MN}/\overline{\text{MX}}$ is tied to VCC (logic high)

8086 itself generates all the bus control signals

**HOLD**

Input signal to the processor from the bus masters as a request to grant the control of the bus.

Usually used by the DMA controller to get the control of the bus.

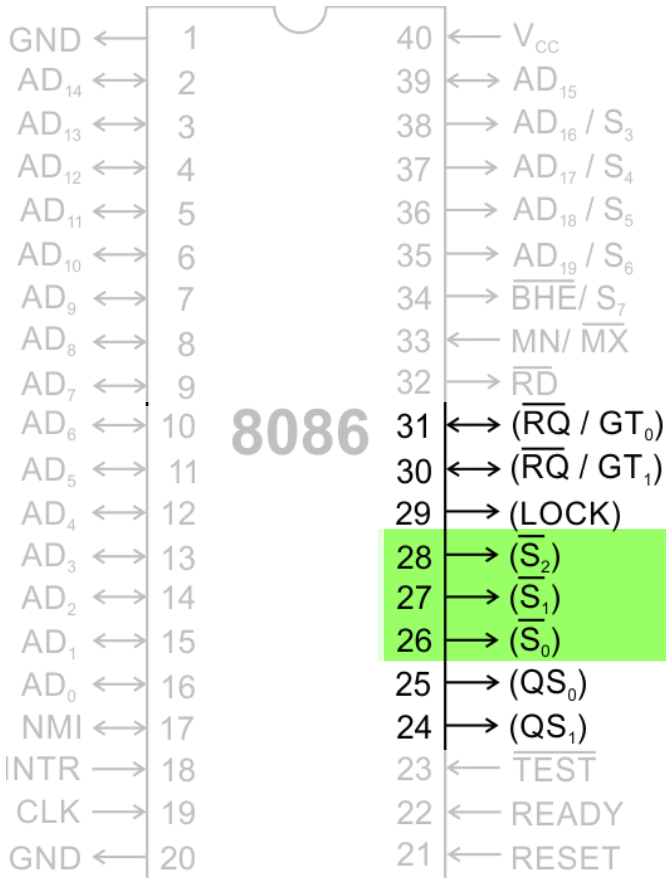
HLDA

(**Hold Acknowledge**) Acknowledge signal by the processor to the bus master requesting the control of the bus through HOLD.

The acknowledge is asserted high, when the processor accepts HOLD.

During maximum mode operation, the $\text{MN}/\overline{\text{MX}}$ is grounded (logic low)

Pins 24 -31 are reassigned

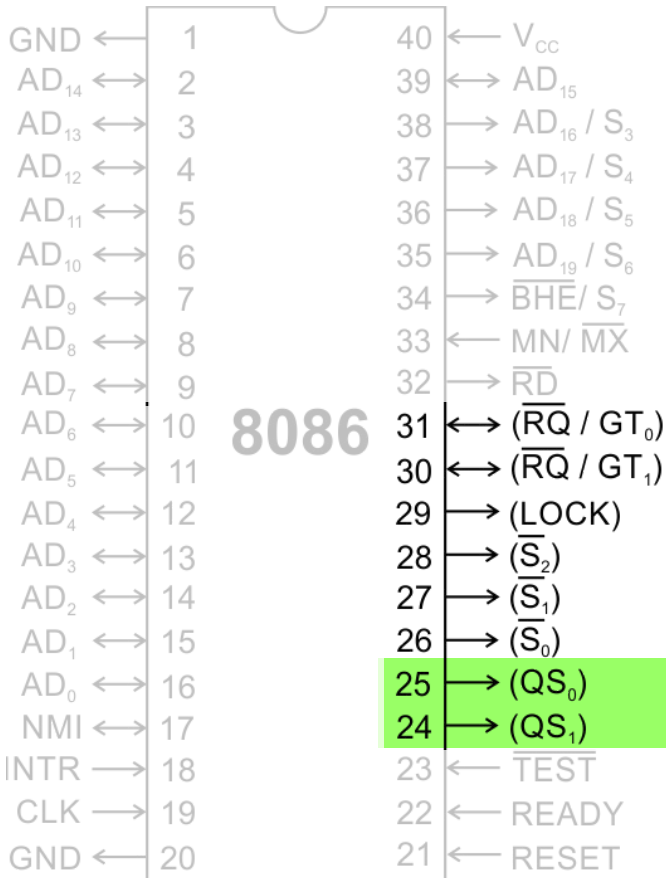


Status signals; used by the 8086 bus controller to generate bus timing and control signals. These are decoded as shown.

S2	S1	S0	Function (Decoded by Bus Controller)
0	0	1	I/O Read
0	1	0	I/O Write
1	0	1	Memory Read
1	1	0	Memory Write
0	0	0	Interrupt Acknowledge

During maximum mode operation, the $\text{MN}/\overline{\text{MX}}$ is grounded (logic low)

Pins 24 -31 are reassigned



(Queue Status) The processor provides the status of queue in these lines.

The queue status can be used by external device to track the internal status of the queue in 8086.

The output on QS_0 and QS_1 can be interpreted as shown in the table.

QS1	QS0	Function
0	0	No operation or queue inactive
0	1	First byte of opcode is in queue
1	0	Queue is empty
1	1	Subsequent bytes of opcode are in queue

During maximum mode operation, the $\text{MN}/\overline{\text{MX}}$ is grounded (logic low)

Pins 24 -31 are reassigned

